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Review article

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Osteoarthritis as a systemic disease

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Abstract

Osteoarthritis (OA) is a painful disorder of the synovial joints characterized by dysfunctional cellular metabolism and extracellular matrix degradation that activates maladaptive repair responses, including pro-inflammatory pathways of innate immunity. OA has historically been viewed as an unavoidable consequence of ageing owing to wear-and-tear of the joint. Most strategies to mitigate OA have primarily focused on the articular cartilage and, more recently, other specific joint tissues, which limits understanding of OA as a systemic disease beyond the joint. Preclinical and clinical data support a paradigm shift in understanding OA as a disease of the whole person, focusing on changes throughout the body that both promote and are caused by OA. This shift provides novel insights into why preclinical research findings have not been successfully translated into clinical therapies. Moreover, a viable strategy for OA drug development could be to treat pain separately from OA structural damage. Challenging classically held views that lack evidence or have been disproven is required for progress. We explore some key emerging questions that should be carefully considered in future OA studies. Adapting and integrating new technologies, techniques, and tools into the rapidly evolving understanding of OA could lead to the development of drugs that might not only treat OA but also provide mechanistic insight into other conditions that result from the dysregulation of systemic factors or from the communication breakdown in inter-organ crosstalk. Such developments would reposition OA researchers from adapting approaches from other fields of science and medicine to leaders in the fight against chronic disease.

Sections

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Key points

- Osteoarthritis (OA) has been viewed historically as an inevitable part of ageing owing to the gradual wear-and-tear of articular joints. Consequently, research and treatment strategies have focused primarily on cartilage or joint tissues.

- Preclinical and clinical evidence indicate that OA is a systemic, whole-body disease, revealing biological changes that both drive and result from OA and clarifying why preclinical discoveries rarely translate clinically.

- OA research progress can occur by challenging outdated assumptions to explore emerging ideas on how OA influences systemic ageing, metabolic dysfunction and cardiovascular health, fostering a deeper understanding of its broader impacts.

- Integrating emerging technologies and analytical tools into OA research will enable the development of therapies that treat OA while uncovering mechanisms underlying systemic dysfunction and impaired inter-organ communication in related chronic diseases.

- This forward-looking approach positions OA researchers to shift from adapting strategies across disciplines to pioneering new methods that could redefine how chronic diseases are studied and treated.

Introduction

Osteoarthritis (OA), a disorder that involves the synovial joints, is a leading cause of pain and disability worldwide. Despite this burden, existing pain management strategies are inadequate[1], which has, in turn, led to musculoskeletal disorders becoming the leading indication for opioid prescription[2]. Furthermore, although OA is the most common form of arthritis, affecting over 600 million people worldwide[3], there are no approved disease-modifying osteoarthritis drugs (DMOADs) that alter the course of joint pathology[1]. Despite musculoskeletal conditions directly leading to morbidity and reduced healthspan, the understanding of OA remains limited. Moreover, the age at which patients are receiving total joint replacement is becoming increasingly younger, further increasing the societal cost of OA[4].

Historically, OA has been narrowly understood as an unavoidable, cartilage-centric, wear-and-tear disease that is inevitable with ageing[5,6]. This narrow perspective has limited the mechanistic understanding of changes beyond the articular cartilage and fails to consider prevalent risk factors, such as injury, obesity or genetics[7], which affect other tissues and organs of the body and co-occur with OA. Approaches to study this painful disease have often focused on evaluating single tissues in isolation (such as cartilage, subchondral bone, bone marrow lesions or synovitis) and assume that OA pathology is focused solely within the joint, or the organ system, which is made up of several joint tissues[6]; however, articular joints do not function in isolation, they are part of a complex organism. Clinical data illustrate that most individuals with OA (59–87%) are managing at least one other chronic condition[8]; for example, people with obesity are far more likely to develop OA[9]. Several studies also indicate that OA disproportionally affects women[10] and individuals from minority groups[11], which contributes to the widening health disparities for OA care, consistent with many other chronic diseases.

In the rheumatology field, the approach to OA is rather unique, as it has historically been studied on an individual anatomic basis, such as a focus on the knee joint or the hand joints in isolation from other joint involvement. This anatomically focused categorization is likely attributed to the focused end-stage treatment of OA, specifically total joint replacement, and the notion that the pathomechanisms of OA may be joint or site-specific. Other rheumatic diseases also have variable joint involvement; however, studies of those diseases do not typically focus only on the most commonly involved joints.

The need for both phenotyping and endotyping in OA is becoming increasingly apparent. Clinical phenotypes might help to identify individuals with OA who are at risk of more severe structural damage, pain or developing comorbidities. Understanding the pathological mechanisms of different endotypes of OA to develop first-in-class therapeutic targets is urgent owing to the exponential increase in disease burden[12] and to facilitate healthy ageing[13].

In this Review, we aim to summarize the current literature to fuel an ongoing strategic shift in the field to consider OA as a systemic, whole-person (or organism) disease, rather than focusing myopically on studying a single joint (such as the knee, hand or hip) or tissue (such as cartilage, bone or synovium) in isolation. First, we state the clinical problem and urgency and summarize the historical view of OA. We share insights from the literature on why preclinical findings have yet to successfully translate to approved clinical therapeutics or improve clinical outcomes. We highlight the ongoing work in the field, such as the 2024 call for a name change to 'systemic OA', which provides evidence of systemic drivers of OA and musculoskeletal pain, and we propose key research questions that the OA and rheumatology research communities might consider in future studies and trials.

Clinical problem and urgency

One of the primary and most recognized risk factors for OA is ageing. Among people with symptomatic OA, the median age at diagnosis is estimated to be 55 years, with 73% of individuals being diagnosed by the age of 65 (ref. 14). However, it is important to note that there are individuals who do not demonstrate knee pain or structural damage with age, indicating that ageing and OA can be distinct processes.

Moreover, other factors also demonstrate strong positive relationships with OA prevalence, including biological sex (that is, sex assigned at birth, throughout the article, 'sex' is used as a shorthand for biological sex). In addition to higher OA prevalence, women also have worse clinical scores than men[10]. In addition, lifetime OA risk is highest in women with obesity and lowest in men without obesity[14], which demonstrates a complex interaction between sex as a biological variable, comorbidities, ageing and OA. Neglecting to account for sex as a biological variable might broaden health disparities, as observed by data on sex differences in anatomy, injury responses and metabolic signatures, which are not simply explained by sex hormones[15,16].

Clinically, a discordance exists between pain and radiographic evidence of structural damage in OA owing to the numerous other factors that contribute to the pain experience[17], and this discordance could be more pronounced in women[10,18]. When people with one painful and one non-painful knee are evaluated, a clear association between structure and pain is observed[19]. When individual joints are considered in hand OA, the joints with structural damage are far more likely to be painful than normal joints[20]; however, when pain severity in the hand is considered with the severity of OA, the association is weak[20]. In the MOST cohort, a pain susceptibility phenotype was identified that increased the risk of developing persistent knee pain over a 2-year period.

This group was characterized primarily by pain sensitization, and these findings suggest that factors beyond pathology in the joint predispose to chronic pain[21]. Together, these data support a complicated relationship between OA pathology and pain but underline the importance of also assessing other factors (such as systemic factors and comorbidities) that contribute to the pain experience.

We and others[22] have proposed an alternative hypothesis to the historical view of OA that obesity-associated systemic metabolic changes are a causal pathogenic factor driving OA[23–27]. Acknowledging that OA is not simply a consequence of increased loading owing to body mass is essential, and we and others have demonstrated that signals from adipose tissue are necessary and sufficient to drive OA in a reversible manner[24,27]. Moreover, loss of muscle[28] and a suboptimal ratio of lean mass to adipose tissue are also associated with increased risk of OA[29]. Muscle loss in an individual with obesity, defined as sarcopenic obesity[30], can increase the risk of OA and worse outcomes[31]. The POMELO study demonstrated that personalized nutrition, resistance exercise and self-management support for people with sarcopenic obesity and OA improved muscle, mobility and quality of life when compared with usual care, focusing on a primary outcome of weight reduction in people with OA[30]. These data suggest that systemic factors can drive OA and, in turn, OA can have systemic consequences (Fig. 1). OA is a serious disease; it increases the prevalence of other chronic diseases that directly reduce healthspan, and contributes to walking disability, which is associated with mortality[26,32,33]. Although evidence suggests that OA might not always directly lead to mortality, the influence of OA on lifespan or healthspan remains to be directly tested. However, these data cannot be obtained if studies are not designed to assess for these parameters; we advocate for a comprehensive view of OA, which incorporates the contribution of multimorbidity and systemic factors, as doing so could provide an important opportunity for meaningful intervention for this increasingly prevalent, painful disease. The compartmentalized, site-specific approach of many OA studies compounded by the historically cartilage-centric focus in OA research, has potentially hindered the opportunity to understand OA as a systemic disease and a systemic driver of disease, or at least the identification of an endotype related to systemic mechanisms.

Endotypes are mechanistically defined as distinct pathways that lead to disease end points[34], which could mean that individuals advance to a similar clinical pain or structural end point through different mechanisms. In the APPROACH trial, investigators demonstrated that longitudinally stable structural disease endotypes might be leveraged to generate biomarker-based endotyping in a clinical trial setting[35]. These data pave the way to both illustrate relationships between systemic factors and OA, and to identify patients who might respond to a particular therapy or intervention. However, treatments that target a particular molecular pathway can only be successful in trials if people with OA who have disease that is driven by that specific pathway are identified for

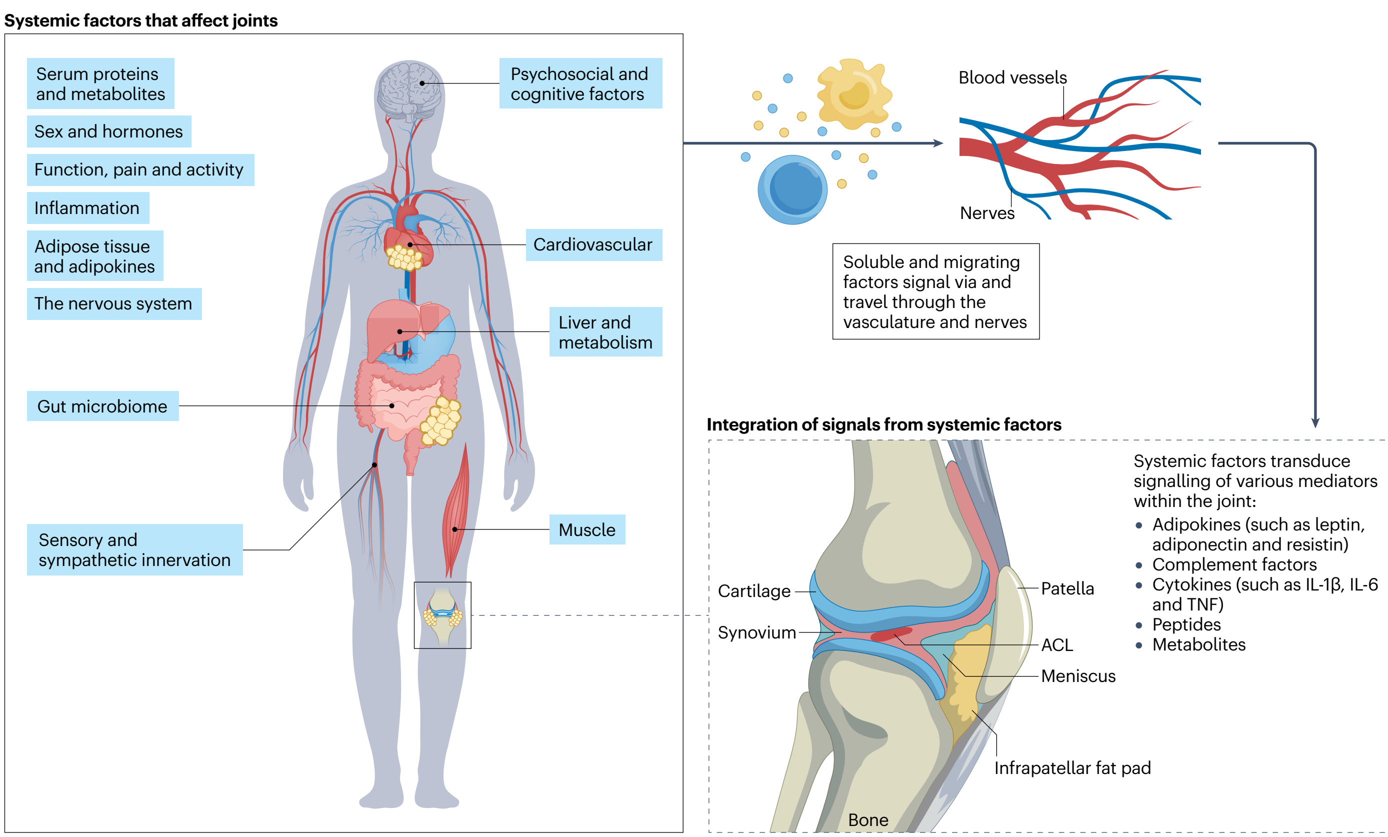


Fig. 1 | Effects of systemic factors on joint health. The precise signals and secreted factors involved in systemic osteoarthritis are unclear. In addition, the tissues and/or cell types in the joint that transduce these systemic signals remain unknown. However, preliminary data indicate that the synovium, blood vessels or bone marrow might be the key transducers of these signals. We posit that a tissue or cell type receives and integrates the signal and subsequently signals to recipient cells, which remain to be determined. Ultimately, these signals drive cartilage loss, bone remodelling, synovitis and other hallmark factors of osteoarthritis. ACL, anterior cruciate ligament.

enrolment into those trials, which demonstrates the need for patient endotyping to maximize treatment efficacy.

Although several biologic drugs have been evaluated in OA, the development of DMOADs has failed in a variety of contexts; for example, in hand and knee OA[36], studies of pharmacological inhibitors of TNF[37–40], IL-6[41], IL-1, matrix metalloproteinases or senolytics have all failed to meet their primary end point[42]. There are many reasons why these drugs have failed (reviewed elsewhere[43–45]), including short duration of treatment and follow-up, enrolling individuals who are too far along in their disease pathology trajectory, and not necessarily including the right patients for the treatment targets being tested[37–42]. Another reason for treatment failure might be because OA is treated as one disease rather than as an umbrella term for a common end pathology that occurs through and is modified by many different pathways, including injury, genetics, sex, obesity, ageing, malalignment, altered biomechanics, inactivity or other reasons[46]. Moreover, renewed excitement and more data to support the hypothesis that OA is a systemic disease are evident in the drugs that are currently pursued for secondary indications for OA, including systemic administration of semaglutide, liraglutide, metformin and other drugs that aim to target systemic hormone or metabolic dysfunction in patients[47].

Historical view of OA

Converging evidence implicates crosstalk among joint tissues, immune pathways and systemic factors such as ageing, obesity and sex-specific biology in driving OA onset and progression. Despite advances in molecular and imaging technologies, translational progress remains hindered by preclinical models that inadequately capture the temporal, cellular and systemic complexity of OA. Innovative approaches (including organoid and 'joint-on-a-chip' platforms, context-relevant models, integrative omics, pain assessments and multi-joint phenotyping) are poised to bridge these gaps and advance a holistic framework that defines OA as a disease of the whole organism rather than of a single joint.

Moving beyond a focus on cartilage damage

OA has been classically viewed as an inevitable condition of ageing, caused purely by mechanical wear-and-tear of the articular cartilage, with little attribution to the role of systemic biological processes in maintaining the overall health of various joint tissues. Early analyses of cartilage were focused on its mechanical function in transmitting joint loads and providing a bearing surface for movement, albeit with a recognition of its lack of intrinsic repair properties[48]. Through the increasing understanding of the role of chondrocytes in the maintenance of cartilage homeostasis, the field has gained an improved understanding of the biological processes that govern the maintenance and health of the joint[49]. Nonetheless, despite the recognition that both biomechanical factors and biological signals (such as inflammation) have a role in OA, the field experienced substantial debate as to the 'tissue of origin' of the disease (that is, bone versus cartilage)[50] and whether joint inflammation was involved in the disease process[51]. Even with the mounting evidence that indicates the involvement of multiple tissue types and both biological and biomechanical processes, 20–30 years ago, investigators were still questioning whether OA was a disease of cartilage or whether it should be termed 'osteoarthrosis' instead of 'osteoarthritis' owing to a lack of classical characteristics of inflammation[52,53].

A role for inflammation in OA

Although historically considered vastly different with independent pathophysiology, there is an increasing appreciation that similar clinical phenotypes occur in OA and rheumatoid arthritis[54], including similar pain and disability scores and involvement of synovial inflammation (albeit synovial inflammation is more florid in rheumatoid arthritis than in OA). Moreover, in many studies, OA serum is used as a control for rheumatoid arthritis serum, which might have skewed the understanding of the relative involvement and pathways of inflammation present in both diseases. Inflammation in OA is readily recognized clinically as joint effusions, but the degree of inflammation differs from other rheumatic diseases[55,56]. Ultrasonography and MRI demonstrate a high prevalence of inflammatory features (such as synovitis)[57], and sustained synovitis is related to considerably higher rates of OA progression[58] and pain[59,60]. Furthermore, changes in the infrapatellar fat pad, consistent with inflammation, are related to patient-reported pain and function in OA, and these relationships differ in women and individuals with obesity[61]. Notably, these signs of inflammation can be observed in patients who have no known history of knee injury[62]. These data are corroborated by single-cell and 'omic atlassing' efforts that aim to use untargeted profiling approaches to demonstrate synovial fluid and infrapatellar fat pad transcriptomic changes consistent with inflammation in OA[63,64].

Owing to this oversimplified understanding and approach to the OA disease process, much of the process of drug discovery for DMOADs has focused on single tissues (cartilage or bone) or single pathways (biomechanical or biological). Historically, drug discovery that targeted single processes, such as cartilage degradation (for example, aggrecanase)[65,66] or bone loss (such as bisphosphonates)[67], showed beneficial effects for bone and cartilage but generally failed in clinical trials. Bisphosphonates, calcitonin or MIV-711 (cathepsin K inhibitor) also showed beneficial effects for cartilage in addition to bone[68] but these drugs also failed.

Although mechanical factors can lead to cartilage loss and bone remodelling, a role for mechanisms beyond tissue wear-and-tear is clear, such as the involvement of important 'mechanobiological' components[69]. Similarly, the biological processes involved in cartilage maintenance are not represented simply by the balance of anabolic and catabolic activities of chondrocytes. Numerous complex pathways of local and systemic inflammation, cell metabolism, ageing, senescence, and innate immune activation interact with both genetic and epigenetic factors in multiple tissues of the joint. This complexity has been shown in other diseases, such as type 2 diabetes mellitus (T2DM) and cardiovascular disease, and is now suggested in the pathophysiology of OA[70]. Taken together, evidence shows that cells and tissues of the joint do not function in isolation but, rather, the joint functions as an organ within the integrated organism; these inter-organ crosstalk interactions are essential in the physiology and pathology of the synovial joint[6]. Another inherent limitation of prior work in OA is that most research is focused on the knee, which does not necessarily translate to other joints (such as the hand and hip).

A vulnerable joint: local and systemic factors drive OA

We posit here the concept that a vulnerable joint might arise from additional systemic predispositions or external driving factors that function in concert to accelerate disease onset and progression. This concept remains to be directly tested. There are several key questions to address; for example, from a clinical perspective, what can be learned about OA as a systemic disease from studying multi-joint pathophysiology? Efforts underscore the value of comprehensive phenotyping of multi-joint OA versus studying OA in single anatomic locations[71]. Another important factor to address is whether unilateral knee OA is the same as bilateral hand OA that involves all proximal interphalangeal joints. The use of animal models with comorbidities, such as obesity and ageing, should be evaluated to determine if they can help to address some of these knowledge gaps. Increasing evidence suggests that the whole-body burden of OA could potentially be sexually dimorphic[15]. A deeper understanding of the pathophysiology of sex differences will be fundamental in advancing patient-relevant outcomes such as functional limitations and quality of life; however, a definition of multi-joint OA is lacking, which limits progress in clinical studies that aim to understand OA as a systemic disease.

The gap between preclinical findings and translation

There are several fundamental mechanistic aspects of preclinical models of OA that require investigation: innervation patterns[72,73]; how the outcomes of interest of the trajectory of OA disease pathogenesis change over time; how obesity, ageing and sex differences alter disease; the influence of genetic and epigenetic risk factors; clear consensus on how to study early signs of disease in mice; and studies that focus on structural damage at predefined study end points. The latter might reflect a lack of time-course studies and the use of models that do not recapitulate key components of OA phenotypes in preclinical models. These gaps require preclinical investigations as well as reverse translation of clinical findings and discovery biology in human tissues. Addressing clinically or translationally meaningful outcomes in these studies is also important.

Although the concept of the joint as an organ is broadly accepted, it remains difficult to address such questions using in vitro models or even in most preclinical models of OA (such as rodents). In vitro models are often limited to the use of a single-cell or tissue type, in which control of the genetic background is limited, and a focused use of either animal or human cells. Many of these limitations are just beginning to be addressed; development of organoids[74,75] and ‘joint-on-a-chip’ models[76–79] are improving, which combine multiple tissue types and potentially allow for control of signalling, directionality and genetically defined human cell types (such as induced pluripotent stem cells) that can be edited to include specific genetic risk factors for OA[80].

Designing preclinical models to interrogate clinically relevant gaps in knowledge

Both in vitro and in vivo models can provide systems whereby genetic risk factors can be defined together with other OA risks (such as obesity, joint injury and ageing). Thus far, such preclinical in vivo models have often been limited by a focus on a single sex (typically, male) with limited outcome measures that integrate all aspects of OA across the body (that is, they focus on a single joint), including measures of pain. Many studies are generally joint-specific without considering multi-joint OA, which is difficult to model preclinically. However, with diet-induced obesity, we demonstrated the presence of multi-joint OA in the knee and shoulder, but not the hip, in Sprague–Dawley rats[81]. A holistic approach in both preclinical and human studies is needed to study all the joints in the body, particularly in preclinical models in which the effect of an inciting agent can be extensively studied, and clinical insights and phenotypes can be reverse translated to probe mechanisms and evaluate treatments. Furthermore, Cre-Lox and transgenic mouse systems that enable several key tissues or cell types to be targeted in the mouse joint (such as the synovium, infrapatellar fat pad and bone marrow) are lacking. Such approaches could be used, to separate the roles of these effector tissues in transducing systemic signals to various joint tissues.

A barrier to gaining foundational understanding regarding endotypes is that it has been common practice to test the mechanisms, safety and efficacy of drugs in young male mice, which contrasts with the clinical OA population that largely comprises women or people from minority backgrounds in the middle-to-late stages of life. The lack of testing in more relevant preclinical models could be because of perceived limitations of animal models; for example, the assertion that female mice do not get OA. Although this statement might be true for histology, in that female mice might demonstrate more modest cartilage damage, data from a recent study showed that, in the destabilization of the medial meniscus (DMM) mouse model, female mice have a similar severity of hyperalgesia to male mice, but the structural changes observed, despite being substantial, were less severe in female mice[82].

Shifting the focus from established OA to early OA in mice

Typically, clinical trials use radiographic assessments of disease, which are relatively insensitive measures. Additionally, many clinical studies focus on studying individuals with established disease and, as such, preclinical assessments have also focused on modelling established or end-stage disease. From the perspective of understanding disease mechanisms, a challenge with studying organisms or people with established disease is that key (and potentially intervenable) risk factors that contribute to the early disease processes are missed. From the perspective of clinical trials, the expectation that targeting a single molecular pathway might be successful is probably unrealistic when the disease is well established, as numerous downstream pathways or factors that influence the outcomes beyond the single molecular pathway being targeted are activated — akin to trying to close the stable door after the horse has already bolted. Disease-related changes will probably occur in people or in model systems a long time before they are studied, and those individuals who have these risk factors but do not develop overt disease could provide important insights into how their tissues, biomarkers or endotypes might differ from those of individuals who do develop disease. In particular, risk factors that lead to disease might persist for many years prior to the development of the established disease. Efforts are under way to develop classification criteria to validate the identification of individuals with early-stage symptomatic knee OA prior to the development of established disease to facilitate epidemiological studies and trials at a stage when both studying risk factors and intervening might be more fruitful[83].

A limitation in the study of early OA is a lack of widely accepted assays or outcomes to identify early signs of disease in preclinical models. Thus far, only histology, which requires a terminal end point, is widely accepted for this purpose but pain hyperalgesia can be detected as early as 2–4 weeks after injury[82,84]. Many single-cell transcriptomic studies of joint tissues in well-established OA have been published. However, the opportunity exists to use a variety of techniques (including measuring proteins and metabolites, using spectral flow cytometry and other untargeted techniques) to profile secreted and circulating factors to define the serum or systemic changes that occur earlier in the disease process since it is well known that animal models of OA generally demonstrate measurable joint damage after 8–12 weeks[85]. Depending on the research question, it could also be valid to use pain outcomes to detect early changes owing to post-traumatic OA independent of structural damage. This focus on uncovering early OA in animal models is important in addressing clinically or translationally meaningful early changes in OA at both the local and systemic levels and in beginning to define acceptable outcomes or disease trajectory benchmarks in well-controlled preclinical model systems.

Systemic metabolic factors in driving early OA

To understand the risk of early OA, younger patients (that is, those individuals who are younger than the typical age of OA diagnosis) should be monitored for risk of OA and pain (that is, earlier than middle age). Although most people aged 30–40 years might not demonstrate overt metabolic distress, the relationship between metabolic changes and OA risk or progression has not been studied systemically to date. A 2025 proteomic atlas of ageing revealed that adipose tissue is among the first tissues to show measurable signs of ageing in the fourth decade of life[86], which can lead to metabolic disturbance that might or might not be monitored clinically as standard of care. These findings emphasize the need to understand the metabolic or systemic comorbidity status of individuals aged 30–40 years to create a deeper understanding of the pathophysiology of early OA, a group that could potentially benefit from interventions to mitigate or prevent OA progression.

Although obesity is defined by body mass index (BMI), lower BMI is not always indicative of a healthy metabolic status and, furthermore, individuals can have an unfavourable ratio of adipose tissue to lean muscle mass despite having a lower BMI. Moreover, measuring body composition and metabolic status, even in people with a BMI of 25–29 kg/m^2, will provide insights into the underlying metabolic risk in younger people[29,87]. Even among those with obesity, there are differential metabolic consequences; for example, sarcopenic obesity or low muscle mass coincident with obesity occur in approximately 10% of individuals with OA, and can influence walking speed, endurance, strength and other patient-reported outcomes[88,89]. The mechanisms of the interactions with obesity, ageing and sex differences that drive OA, genetics and epigenetics, among other key comorbidities with systemic manifestations, remain to be clarified. Indeed, physical function is a better prognostic marker for premature mortality than almost any laboratory test among those with or without a form of arthritis, obesity or comorbidities[90–96].

Interrogating pain and behaviour in animal models

Pain is by definition a complicated construct involving both sensation and emotion, articulated by the International Association for the Study of Pain as “an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage”[97]. There are strong associations and a probable bidirectional relationship between emotional and cognitive factors (such as depression, anxiety, pain catastrophizing and self-efficacy), sleep and comorbidities; for example, people with a high burden of psychological factors, comorbidities and sensitization but a low degree of OA report the highest level of pain[24]. In these people, targeting cartilage may not improve their pain, yet the outcomes of interest in preclinical studies are almost exclusively focused on mitigating, reversing or preventing joint damage, and clinical trials do not differentiate inclusion criteria based upon those factors that contribute to the pain experience of an individual. Although some studies describe factors that might have separable, and even paradoxical, roles in driving pain and structural damage[24], historically, factors do not seem to be pursued or reported if an element of structural damage was not involved; the outcome of this is that several relevant or interesting targets have not been pursued. This lack of reporting might be because pain has only been considered in preclinical models in the past few years[98].

Notably, there are largely three categories of pain that are present in rheumatic diseases: nociceptive pain, neuropathic pain and nociplastic pain, which are reviewed elsewhere[99]. In addition to nociceptive pain (caused by nociceptive input from pathological peripheral tissues), nociplastic pain, which is characterized by altered nociceptive processing and central sensitization in the absence of tissue pathology, is also present in some people with OA, and is probably more prevalent in people with OA and comorbidities[100]. The identification of factors involved in different pain mechanisms can improve understanding of the separable and overlapping contributors to pain and structural damage in OA and of how these might be driven or affected by systemic factors[24,82] to identify new promising therapeutic targets in preclinical and clinical studies. As studies do not often consider the whole joint or the whole organism, many preclinical models do not capture potential key aspects of relevant OA pain mechanisms that can be affected by systemic factors (such as diet, sex, environment, genetics, epigenetics and the microbiome)[25] and the interactions of these relevant factors. Notably, knee pain is not relieved by joint replacement in ~20–30% of individuals, probably owing to persistent pain sensitization and other systemic contributors to pain beyond joint pathology. More data are needed to better understand this relationship.

Factors beyond the joint

Genetic, metabolic and systemic factors outside the joint converge to drive OA structural damage and symptoms. Preclinical and clinical data highlight roles for adipokines, inter-organ crosstalk and sex-specific mechanisms that might be affected by age and comorbidity status. Targeting metabolic dysfunction, inflammation and systemic pathways offers promising avenues for disease-modifying therapies.

Genetic risk

In the past three decades, data from large-scale genome-wide association studies (GWAS) have revealed genetic variants of relevant functional consequences in hand OA. These variants have testable treatment targets, such as Matrix Gla protein (MGP; a vitamin K-dependent protein) and aldehyde dehydrogenase 1 family A2 (ALDH1A2; a key enzyme for the synthesis of all-trans retinoic acid), both of which can be targeted with available inhibitors[101]. Such insights further support the concept of OA as a systemic disease; for example, even post-traumatic knee OA, which is typically considered to be isolated to the affected joint, could potentially have an underlying genetic predisposition or systemic driver to explain why some people go on to develop OA but others do not despite the same mechanism of injury (such as severity of an anterior cruciate ligament tear). Indeed, there is some evidence of a genetic contribution to the severity of post-traumatic OA[102,103]. At the same time, these studies highlight the multifactorial nature of OA since not all individuals with these genetic variants develop OA. The Genetics of Osteoarthritis consortium is undertaking work on GWAS of OA endophenotypes[104] that will provide foundational mechanistic understanding of the role of genetic drivers of OA, which have been demonstrated, in part, to be associated with systemic metabolic diseases[105]. The roles of epigenetics and gene regulatory mechanisms in driving OA are also being explored in addition to genetic drivers, which have been reviewed elsewhere[106–108]. Some of these GWAS studies have been functionalized in preclinical models to assess mechanistic gene regulatory interactions; for example, the *R2de* region of the mouse *Gdf5* gene, which is among the most cited OA risk variants[109]. Genetic testing services demonstrate how single-nucleotide polymorphism analysis could become a more accessible tool that could be used to gain a deeper mechanistic understanding of the causal role of genetics in OA and thus provide potential treatment targets.

Reverse translation of clinical phenotypes and endotypes

One opportunity to bridge the gap between preclinical models and clinical data is the use of preclinical models that can demonstrate the necessity and sufficiency of factors of interest such as leptin[110–113] (Box 1). Mice or other preclinical models enable more comprehensive profiling that can help to establish cause and effect, which could enable better translation if the models were more fit for purpose. The association between obesity and symptomatic OA is stronger than between obesity and radiographic OA[114], and adipose tissue exerts effects on OA-related pain independent of body mass[24,25,27]. Quantifying adipose mass, body mass and obesity is challenging, and clinical data on the role of adipokines are conflicting. However, the systemic effects of obesity might be explained by leptin, which has been shown to have promising links to pain, radiographic changes and American College of Rheumatology diagnostic criteria. Leptin, a satiety hormone, is the most consistently increased adipose tissue-derived mediator reported in obesity-induced OA[115,116]. Leptin-knockout mice are protected from OA, despite being morbidly obese[117,118], and leptin is sufficient to induce OA in the absence of adipose tissue in mice[24]. However, the precise mechanistic role of leptin, its target tissues in the joint and elsewhere, remains unknown. In a cohort of people with hand OA, individuals with overweight and obesity reported more severe pain not only in the knees and hips but also in their hands[114,119]. The association with hand pain was mediated through leptin, supporting the systemic effects of obesity on pain, which have also been recapitulated in mouse models of knee OA[24].

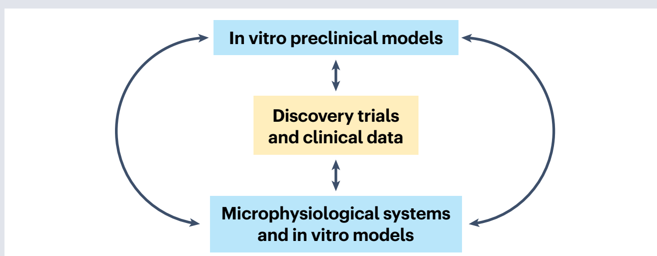
Similarly, visceral adiposity, but not subcutaneous adipose tissue, has been associated with knee pain severity[120], highlighting the importance of differentiating metabolically active adipose tissue from overall body mass. The effects of obesity on hand pain and painful total body joint count in this study were partially mediated by leptin and high-sensitivity C-reactive protein (CRP)[120]. Furthermore, bariatric surgery not only reduces lower extremity joint pain but also hand pain and reduces sensitization[121,122]. Coincidentally, decreases in leptin and increases in glucagon-like peptide 1 (GLP1) are observed almost immediately post-operatively, even before weight loss[123], which suggests that altering systemic factors and mitigating pain can be separate from overt weight loss. There are also new spatial tools to consider for OA research (including spatial profiling tools such as spatial transcriptomics, spatial metabolomics and spatial proteomics) that we and others have begun to develop for mouse models and human studies[63,64,120], which will provide important missing spatial context on how the joint responds to systemic factors in both human and preclinical discovery studies. These spatial tools help to overcome the limitations of the current strategies used to target tissue-specific changes or observe them in situ or in a whole-joint preparation (that is, for samples obtained from mice, human cells or organoids).

Inter-organ crosstalk between adipose tissue and OA

Research has demonstrated a causal relationship between adipose tissue and knee joint damage in mice[27,124], which is corroborated by

Box 1 | Forward and reverse translation in OA research

To advance the understanding of osteoarthritis (OA) as a systemic disease, in vitro, preclinical and clinical samples can be used in a forward and reverse translational capacity. All of these approaches bring opportunities to OA research:



Microphysiological systems and in vitro models

- Holistic phenotyping (tissues, fluids (such as serum and synovial fluid), whole joints and multi-joints)
- Controlled systems for early OA assay development
- Measuring pain and structural changes in vitro
- Drug repurposing
- Inclusion of clinically relevant factors such as sex, obesity, ageing and comorbidities
- Time-course studies
- Studies that test the directionality of signalling

In vivo preclincial models

- Use of human materials with known risk factors
- Assessing inter-organ crosstalk
- The ability to establish cause and effect
- Inclusion of clinically relevant factors such as sex, obesity, ageing and comorbidities
- Time-course studies
- Assessment of pain factors
- Development of early diagnostic tools in mice to better study early OA

Discovery trials and clinical samples

- Holistic phenotyping (tissues, fluids (such as serum and synovial fluid), whole joints and multi-joints
- Measuring pain and structural changes to probe both aspects together and separately
- Determine the timing and type of rare OA signs and symptoms
- Inclusion of clinically relevant factors such as sex, obesity, ageing and comorbidities
- Time-course and prospective cohort studies
- Investigating early OA in people at risk
- Determining factors that drive persistent pain after treatment

Glossary

Deep learning
A subset of machine learning that uses multilayered artificial neural networks to model complex data patterns.

Elastic net
A regularization method used in machine learning that combines Ridge and Lasso regression models.

Healthspan
The number of years lived in good health, free from chronic disease or disability.

Lifespan
The total number of years that a person lives (from birth to death).

Neuropathic pain
Pain caused by lesions, injury or disease that affects the somatosensory nervous system.

Nociceptive pain
Pain that occurs owing to the activation of nociceptors due to tissue injury, inflammation or disease.

Nociplastic pain
Pain originating from altered nociception without clear evidence of tissue damage or nerve injury.

Pain catastrophizing
A pain-specific response that involves magnified negative thoughts and feelings of helplessness.

clinical evidence[100,125,126]. Moreover, adipose tissue has been suggested to be essential to organismal ageing. Thus, monitoring individuals with metabolic risk factors for OA-related pain might be important to better understand, separate and triangulate the individual and the combined impact of multimorbidity, adipose tissue, and metabolic disturbance on ageing and age-related musculoskeletal disease[100,127]. For example, the role of fat in OA has been shown using a mouse model of lipodystrophy[24,27]. Global unsupervised multiomics of conditioned media from adipose tissue implants revealed that leptin exerts a regulatory effect on adipsin (also known as complement factor D), which is part of the alternative complement pathway[16]. Interestingly, the loss of adipsin protected OA joints but resulted in pronounced sensitivity to pressure-pain hyperalgesia, which was detectable as soon as 2 weeks after injury in both male and female mice[82]. Evaluating and understanding the role of the innate immune system and other factors that might be activated or upregulated by adipokines, such as leptin and adipsin, and have a unique role in pain and/or structural damage, might be foundational data from which new drugs could be developed[24,27,82].

Repurposing anti-obesity and metabolic dysfunction drugs for OA

Much excitement has come from the findings of the STEP 9 trial of semaglutide, a GLP1 receptor agonist, which demonstrated greater pain improvement and weight loss in the intervention arm than in the placebo arm in patients with OA[26]. Similarly, GLP1 receptor expression has been reported in joint tissues in rodents, and treatment with systemic liraglutide in rodents alleviates pain, independent of weight loss[128]. Although the GLP1 family of drugs has been proposed to slow down the ageing processes and improve healthspan[129], the mechanism by which this occurs and how it potentially improves OA specifically is unknown. Some studies report negative outcomes of GLP1 drugs, for example, worsening of OA[130,131] and bone loss[132]. Another prominent concern is loss of lean muscle mass[131,133] in concert with adipose tissue to achieve overall weight loss, resulting in a potential risk of sarcopenia and detriment to strength and function. These findings indicate that understanding the systemic mechanisms of OA is essential to better use these incretin-targeting drugs and identify patients who will benefit most from their administration. This point is especially concerning as an increased risk of OA has been reported in individuals with muscle loss or weakness[31,88,134].

Other potential therapeutic targets to treat metabolic dysfunction could be identified through new insights gained from studies of the gut microbiome. Although initial studies aimed to understand the role of the gut microbiome as a link between diet and OA[135–138], data now indicate that the gut microbiota–GLP1 axis could be a highly attractive drug target for OA[47]. Several association studies have linked gut microbiome dysbiosis to OA[139,140], and DMM-induced OA is reduced in mice that lack a gut microbiome[141]. Faecal microbiota transplantation can confer resilience in mouse models of OA, which suggests that microbiome manipulation or transplantation might be a viable therapy for OA[142]. It remains unclear if the microbiome-mediated influences on the joint occur in a manner dependent or independent of lipopolysaccharide signalling[135,143], but research is ongoing[144]. Data from the Lifelines-DEEP study demonstrate that this association is driven by local inflammation in the knee joint[145]. Further details about the pathophysiological role of the microbiome in OA, the role of the microbiome in systemic OA and the therapeutic potential of targeting the gut microbiome have been reviewed elsewhere[146].

Another promising therapeutic approach is targeting insulin resistance systemically. Treating the metabolic effects that might drive OA using metformin, a T2DM drug, has been shown to mitigate structural OA pathogenesis, pain and immune changes in mouse models and people with OA[147]. A high number of individuals with OA are at risk of or diagnosed with T2DM or metabolic syndrome. In a 2025 trial of people with painful knee OA and a BMI of ≥25 kg/m^2, metformin was superior to placebo for pain reduction[148]. Metformin is also currently being interrogated as a candidate anti-ageing drug, as it was shown to slow the effects of ageing in the TAME trial[149,150], and is being evaluated in OA in a clinical trial aiming to prevent post-anterior cruciate ligament injury OA (that is, post-traumatic knee OA; NCT06096259). These trials provide further support for the notion that a complex interaction between metabolism and ageing promotes OA pathogenesis.

Sex hormones and sexual dimorphism in OA

Another key factor at the interface of systemic factors in OA, ageing and adipose tissue is menopause. A sizable proportion of individuals with OA are women in post-menopause with obesity, and the female sex is one of the strongest predictors of OA development[151]. This observation underscores the need to better understand the role of sex, hormones, pain and metabolic dysregulation in OA. Data on the effects of hormone replacement therapy (HRT) on OA and knee pain are mixed. In a post hoc analysis of a large trial of HRT in women with cardiovascular disease[152], there was no difference in knee pain compared with placebo. More trials are needed to better understand the potential role of HRT in OA pain[153,154].

Xist (X-inactive specific transcript) is a non-coding RNA that has a key role in the X-inactivation process. In 2024, Xist ribonucleoproteins were shown to drive innate immunity in women[155], which is highly relevant to OA pathogenesis and might be another mechanism by which sexually dimorphic systemic phenotypes are driven in OA. Novel models to study sexual dimorphisms can be used to understand the role of cell-intrinsic and systemic effects of chromosomal and gonadal sex on OA and pain but more work is needed[16,156].

Although much interest has been focused on hormonal differences as an explanation for the high prevalence of OA in women, studies thus far have not yielded actionable insights. For example, sexual dimorphism is evident in cartilage before menopause and can be affected by chromosomal sex. Dimorphic phenotypes are not reversed with ovariectomy, and ovariectomy does not fully replicate the natural progression of OA in women in post-menopause. Future studies could consider investigating dimorphic factors that are not driven by sex hormones or cell-intrinsic effects of sex. For example, measuring dimorphic circulating factors, such as proteins, gene transcripts and metabolites, in the serum and designing studies to determine how these dimorphic factors are affected by obesity, age, and other known risk factors for OA could contribute to the understanding of the complex role of sex as a biological variable in systemic OA.

Repurposing DMARDs as DMOADs

DMARDs are successful treatments for many rheumatic diseases, and given some shared potential pro-inflammatory drivers of rheumatic diseases with OA, there is potential to repurpose approved drugs for OA indications with appropriate data and trials. Inflammation is another key link between OA and systemic drivers. Several phase II trials of inhibitors of IL-1β (a key cytokine in OA) have either shown negative results or demonstrated a small effect that might not be clinically relevant or translationally meaningful. Similarly, several trials of colchicine, a drug with anti-inflammatory properties, in knee and hand OA have been negative. A commonality across these trials is that they have been small in size and of short duration (for example, of 6 months). The CANTOS trial, a large placebo-controlled trial of canakinumab, an IL-1β inhibitor, was conducted in ~10,000 people with cardiovascular disease and an increased level of high-sensitivity CRP, followed for a median of 3.7 years. In a post hoc analysis, those on canakinumab had a ~40% lower risk of joint replacement compared with placebo[157]. Similarly, the LoDoCo2 trial of colchicine in ~5,500 people with coronary artery disease (followed for a median of 2.4 years), demonstrated a ~30% lower risk of joint replacement compared with placebo[158]. Since knee replacement surgery is predominantly performed for knee OA, it necessarily implies that the lower rates of joint replacement in the anti-inflammatory treatment arms were related to lower knee OA structural progression, symptoms, or both, and highlights the relevance of considering a systemically acting drug capable of inhibiting inflammation in the whole body or organism. These cardiovascular trials also highlight the challenges with OA trials thus far, which have probably been too small in size, of too short a duration and too heterogeneous to identify an efficacy signal. Efforts must now be made to understand which patients are most likely to benefit from such treatments, again reiterating the importance of systemic patient phenotyping.

Future research opportunities

The paradigm shift towards regarding OA as a potential whole-body disease that leads to pain and loss of physical function, at least in a subset of patients, is a useful lens through which new insights into OA can be gained. However, identifying those factors that limit more representative clinical and/or translational studies is an important step. Some factors include sample size, heterogeneity of study samples

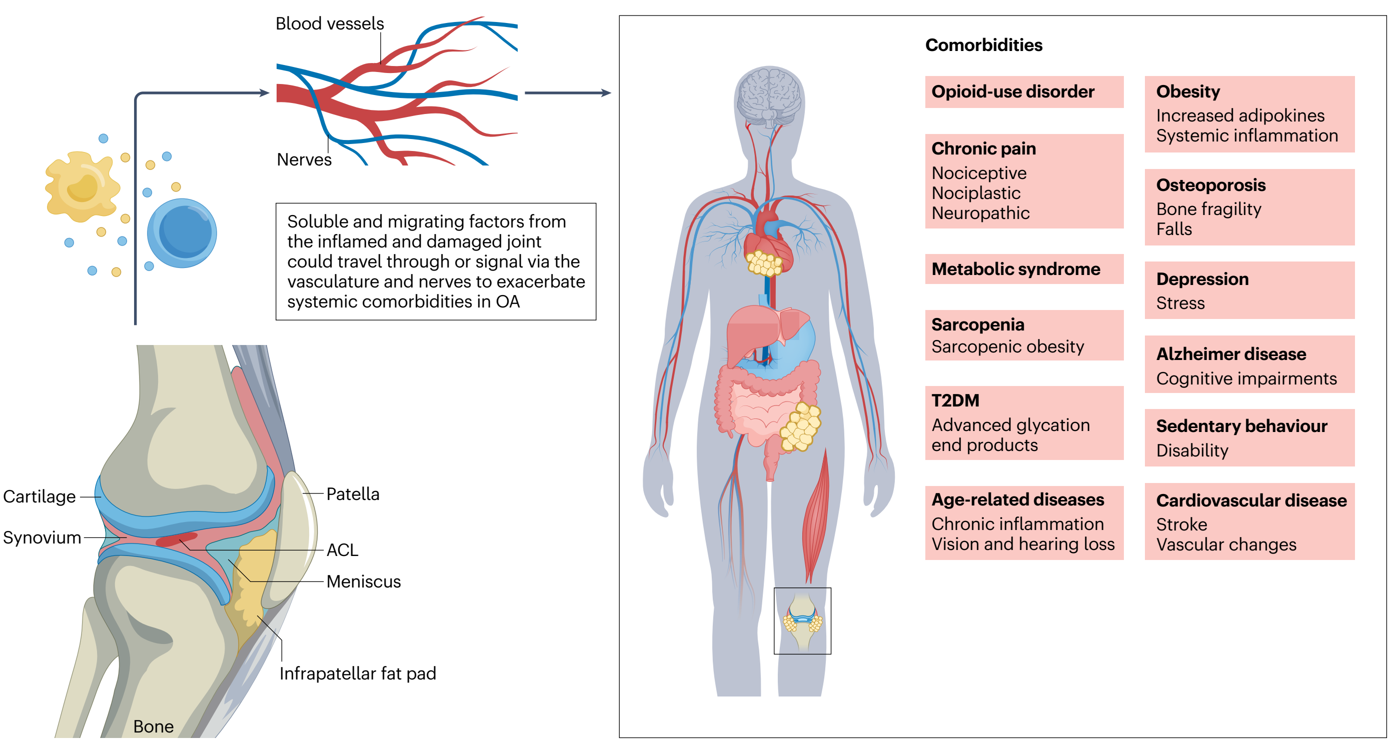


Fig. 2 | Effects of joint health on systemic factors and comorbidities. Although data that establish cause and effect remain to be generated, changes to the local joint environment are coincident, if not drivers, of systemic pathology. Several studies indicate associations with the onset and progression of several diseases and disorders that are systemic in nature (shown in red boxes). It remains unknown if the driving factors (genetics, epigenetics, environment, comorbidities, sex and age) for these systemic diseases also promote osteoarthritis (OA), or if OA promotes these factors independently. ACL, anterior cruciate ligament; T2DM, type 2 diabetes mellitus.

Box 2 | Questions and directions for future OA research

1. Is osteoarthritis (OA) a systemic disease that generates symptoms and pathologies in the joint?
2. Which joint cell populations are vulnerable to systemic or adipose tissue-derived signals, and which populations might drive systemic changes?
3. What is the relationship between pain and structure, and what is lost by focusing on one or the other and not both?
4. How does the whole organism communicate with the joint, and how can clinical observations be reverse translated to better model whole-organism disease using preclinical or in vitro approaches?
5. Can failed trials or interventions be leveraged to better understand potential systemic mechanisms of OA?
6. What can be learned through the integration of existing datasets and cohorts to mine for meaningful clues about the clinical problem of OA?
7. Can OA be used as a model to better understand other diseases of multi-organ ageing?
8. Why do some people get OA with specific risk factors (such as knee injury) and others do not, and how can the 'multi-hit hypothesis' be best evaluated given that joint injury often but not always causes OA, and odds increase with systemic effects such as obesity?

and sample bias, and the fact that discovery research is not typically conducted on systemic factors, largely owing to the limitations of available profiling tools in past efforts. Objective, untargeted profiling of a variety of circulating factors (such as proteins, metabolites[159], cells and immune populations) and multiomic changes in an unbiased manner in blood (that is, serum or plasma) and synovial fluid[160] in early OA is an exciting opportunity for future studies[161]. These data can be objectively integrated, assessed and analysed using machine learning methods such as deep learning[162] and elastic net[163].

OA as a driver of systemic disease

In addition to the concept that systemic factors can influence OA, there is growing evidence that shows the effects of knee pain and damage on the rest of the organism, illustrating a bidirectional relationship. This concept has also been demonstrated in a study that showed that knee injury and pain might cause changes to the cardiovascular system[164], brain[165,166], adipose tissue[24] and other organ systems (Fig. 2). However, observational data from systemic metabolic diseases have not demonstrated consistent associations or shown the same findings as in preclinical models. The potential reasons for this include preclinical models not being fit for purpose (many animal models have limitations in recapitulating human OA); the timing of studying people with well-established disease being too late to understand mechanisms that probably started decades earlier; and the fact that available tools to study systemic effects in humans, typically blood biomarkers, which are often studied in isolation of one another at a single time-point in people with well-established disease, are inadequate to gain insights regarding relevant exposures in the joint environment (and local microenvironment within the joint) at the time of disease initiation. Spatial high-resolution tools[24,108,161,162], such as spatial transcriptomics[64], spatial proteomics[167] and spatial metabolomics[168] and whole-joint anatomical mapping efforts (REJOIN study)[73], will help to triangulate how cells in joint tissues respond to external signals and how cell-intrinsic responses might dictate protection or vulnerability in different tissues. This tissue-specific response might also manifest as an endotype.

A need for unbiased discovery

Reimagining discovery approaches to integrate basic and clinical studies will improve the understanding of mechanisms and targets for systemic OA. To this end, future studies should consider using high-dimensional multimodal assays to analyse human samples for discovery and validate them in preclinical models with the appropriate controls, which could provide new insights. This approach contrasts with performing discovery studies in rodents and designing translational and clinical studies from preclinical mechanistic data, although both approaches are important. The use of an integrated, untargeted multiomics approach to analyse serum and synovial fluid samples might begin to provide much-needed, meaningful insights into systemic OA pathophysiology. Future work should continue to examine the roles of systemic factors on the joint by comprehensively investigating risk factors, the role of tissues outside the joint (tendons, muscle, adipose, cardiovascular system and the nervous system), and the effects of inter-organ crosstalk and systemic mediators (Box 2).

Conclusion

Collectively, these findings underscore the complex and intertwined relationship between systemic factors that drive OA and pain. A shift towards considering systemic drivers of OA will enable the identification of other druggable targets that can be used to treat, manage or prevent OA as well as a myriad of devastating interrelated diseases that co-occur with ageing and obesity. This approach would position the OA community as leading the way for the development of novel treatment strategies for metabolic disease and ageing. Such profiling, for example, of blood, synovial fluid or other systemic factors, might also improve clinical triaging of patients and better identify patients at risk owing to multimorbidity. There remains a key role for preclinical models and in vitro studies to pave the way towards a mechanistic understanding of systemic factors that drive OA, and how OA might promote changes in systemic factors that lead to multimorbidity. Pain assessments, earlier evaluation, reverse translation of clinical paradigms and systemic profiling will provide a deeper, much-needed understanding of the systemic OA phenotype. This ongoing paradigm shift highlights the opportunity to consider the whole patient and pain when evaluating potential individuals with OA or those at high risk of disease in the future.

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